

Part1)

Question 1)

In DBMSs, data is managed centrally in a database (which minimizes redundancy and inconsistency), whereas in traditional file systems, data is stored in separate files, which often leads to duplicate data and inconsistencies.

DBMSs support security and integrity constraints. Traditional file system has limited security and no built-in integrity enforcement

Scenario: A university student registration system.

A DBMS is the better choice because multiple departments (admissions, acads, finance, hostel) need to access the same student data simultaneously. The DBMS ensures consistency, prevents duplicate records, and supports concurrent access.

Question 2)

Strong entity has its own primary key and can be identified without depending on another entity
Weak entity does not have a complete primary key of its own and depends on a strong entity for identification.

The primary key is formed by combining:

- 1)The primary key of the owner entity, and
- 2)The weak entity's partial key (discriminator).

The relationship connecting a weak entity to its owner entity is called an Identifying Relationship.

Question 3)

The phone number should be modeled as a multivalued attribute because a student can have multiple phone numbers. since a simple attribute allows only one value, and a composite attribute is used when an attribute can be divided into components, which is not the case above

Question 4)

Super Key:

A set of one or more attributes that uniquely identifies each tuple in a relation.

Candidate Key:

A minimal super key, meaning no attribute can be removed while still maintaining uniqueness.

Candidate -{roll no},{email}

Super key- {roll_no, name} , {email, branch}

Question 5)

The referential integrity constraint is violated. This constraint prevents records from referencing non-existent rows in another table. The insert operation should be rejected and the DBMS should report a foreign key constraint violation error. This ensures database consistency and prevents orphan records.

Question 6)

- 1) The additional structure that must be created is a separate relation (junction/associative table). eg: Student(StudentID, Name)
- 2) The primary key consists of the combination of the primary keys of the participating entities.
- 3) In an M:N relationship:
One student can enroll in many courses.
One course can have many students.

If CourseID were stored directly in Student: A student would need multiple CourseID values and vice versa. This violates the principles of the relational model (attributes should contain atomic values). Therefore, a separate associative table is required to efficiently and correctly represent all combinations.

Part2)

Section A)

Customer

Attributes: CustomerID (Key), Name, Email

Relationship: Places Order (1:N)

Justification: A single customer can place multiple orders, but each individual order belongs to exactly one customer.

Restaurant

Attributes: RestaurantID (Key), Name, Location

Relationship 1: Receives Order (1:N)

Justification: A restaurant can fulfil multiple orders, but each order is sent to exactly one restaurant.

Relationship 2: Offers MenuItem (1:N)

Justification: A restaurant may host many menu items, but a specific menu item record is unique to that restaurant.

Order

Attributes: OrderID (Key), OrderDate, TotalAmount

Relationship: Contains MenuItem (M:N)

Justification: An order can contain multiple menu items, and the same menu item can be requested across many different orders.

Section B)

Multivalued Attribute: Phone Numbers (Customer)

this because a single customer can have multiple distinct phone numbers at the same time.

Composite Attribute: Address (Customer/Restaurant)

An address can be broken down into smaller, meaningful sub-attributes, such as street, city, and pin-code.

Section C)

Schema Notation

OrderItem(OrderID, ItemID, Quantity)

Primary Key: Composite key of (OrderID, ItemID)

Foreign Keys:

OrderID references Order(OrderID)

ItemID references MenuItem(ItemID)

Additional Attributes: Quantity (tracks how many units of the specific menu item were ordered)

Part3)

